Arnav Gupta

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About

Independent NLP/ML researcher based in Nepal. Research interests span multi-document summarization, faithfulness evaluation, iterative refinement pipelines, and feature selection. Broadly interested in mathematics and physics — particularly calculus, differential equations, linear algebra, and classical mechanics — and in existentialist philosophy.

Education

Sagarmatha Secondary School — **Bandipur, Siraha, Nepal**
Grade 10 | GPA: 3.87 / 4.00 (Outstanding) — *Expected SEE: 2026*

Experience

Suvidha Foundation — **September 2025 – November 2025**
Machine Learning Intern — *Remote*

- Researched multi-document abstractive summarization with transformers under mentorship of a PhD in NLP.
- Built proof-of-concept summarization system with hierarchical context aggregation.
- Participated in community outreach programs supporting underserved populations.

Publications & Preprints

Verifier Exploitation in NLI-Guided Iterative Refinement: A Controlled Empirical Analysis

- Arnav Gupta. Independent Researcher, Nepal. Preprint submitted to arXiv (on hold), 2026. github.com/MrArnav69/AnchorSum
- First controlled empirical documentation of verifier exploitation in zero-gradient, prompt-only iterative refinement: a second revision cycle inflated SummaCConv by +0.185 while collapsing BARTScore$_{\mathrm{s}\to\mathrm{d}}$ by −2.566 nats, confirmed by Wilcoxon W=0 (p=2.68×10^{-83}) across all 498 instances without exception.
- Identified truncation-exploiting content removal as the exploitation mechanism; formalized a three-condition, annotation-free detection protocol applicable to any NLI-guided iterative refinement pipeline.
- AnchorSum achieves a 6.3% relative SummaCConv inconsistency reduction over 498 Multi-News instances (p=4.49×10^{-28}); outperforms fine-tuned baselines (BART, PEGASUS, PRIMERA) across all LLM-as-judge dimensions in a dual-judge jury evaluation.

Research

Independent Researcher — **2025 – Present**
Machine Learning & Natural Language Processing

- Conduct independent research in machine learning and natural language processing, with focus areas spanning multi-document summarization, faithfulness evaluation, iterative refinement pipelines, feature selection, and modular inference-time systems.
- Designed AnchorSum, a modular training-free multi-document summarization pipeline combining entity-guided anchor extraction, anchor-conditioned generation, and dual-mode NLI/entity faithfulness auditing.
- Empirically documented verifier exploitation in prompt-only iterative refinement and proposed a multi-metric triangulation protocol for its detection — establishing that the failure mode is a property of feedback structure, not optimization mechanism.
- Outperformed fine-tuned encoder-decoder baselines (BART, PEGASUS, PRIMERA) across all LLM-as-judge evaluation dimensions using a zero-shot inference-time pipeline with no parameter updates.

Projects

Transformer from Scratch | *Python, PyTorch, NumPy* | GitHub — **2025**

- Implemented the full Transformer encoder architecture from first principles: sinusoidal positional encoding, multi-head self-attention with scaled dot-product, position-wise feed-forward networks, residual connections, and layer normalization.
- Trained on a synthetic cyclic next-token prediction task demonstrating end-to-end autoregressive sequence modeling without reliance on high-level libraries; structured as a modular Jupyter notebook bridging mathematical derivations and working PyTorch code.

House Price Prediction | *Python, Scikit-learn, XGBoost, Pandas, NumPy* | GitHub — **2025**

- Built a regression pipeline on the Ames Housing dataset (Kaggle) covering EDA, data cleaning, and feature engineering — including composite features (TotalSF, TotalBathrooms, PorchSF), age features, log transforms, and ordinal encoding.

- Compared Ridge Regression (baseline, CV $RMSE_{log}$ 0.1445) against tuned XGBoost (CV $RMSE_{log}$ 0.1229, validation RMSE $26,847); hyperparameters optimized via GridSearchCV with cross-validation and residual analysis.

10D Hypercube Visualization | *Python, NumPy, Scikit-learn, Matplotlib* | GitHub **2026**

- Constructed and visualized a 10-dimensional hypercube (1,024 vertices; 5,120 edges) using PCA (2D and 3D projections), t-SNE non-linear manifold embedding, and layer-stratified views grouping vertices by Hamming weight.
- Generated comparative visualization panels across all methods with edges color-coded by the varying dimension, illustrating how geometric structure is preserved or distorted under each projection.

Live Emotion Detection | *Python, DeepFace, OpenCV, TensorFlow, MTCNN* | GitHub **2025**

- Built a real-time facial emotion classifier using DeepFace with MTCNN-based face detection, supporting 7-class emotion recognition (FER2013) on live webcam feed across multiple simultaneous faces.
- Implemented temporal smoothing via a 10-frame sliding window to stabilize predictions, processing every third frame for real-time throughput; overlaid color-coded confidence bars and an FPS counter.

Writing

Medium | medium.com/@mrarnav69 | *ILLUMINATION & Independent*

- The Real Productivity Hack Nobody Talks About: Mastering Task Switching — Published in ILLUMINATION (2026)
- Your Productivity Problem Isn't Focus — It's Leisure (2026)
- Understanding the Transformer Architecture from 'Attention Is All You Need' (2025) — part of an ongoing ML paper summary series also shared on LinkedIn

Programs & Fellowships

Harvard Computer Society AI Bootcamp **2025**
Artificial Intelligence Fellow *Remote*

- Completed intensive AI program collaborating with Harvard students, focusing on frontier AI systems and real-world deployment challenges.
- Studied Transformers in depth and implemented one from scratch; covered embeddings, positional encoding, self-attention, cross-attention, and feed-forward networks.
- Explored Reinforcement Learning: MDPs, Value Iteration, Policy Iteration, SARSA, Q-Learning, and Multi-Armed Bandits.
- Analyzed LLM architectures (ChatGPT, Claude, Gemini, Perplexity) and researched hallucination mitigation strategies.
- Completed hands-on projects: fine-tuned BERT, built a mini-LLM, implemented Vision Transformers, and constructed neural networks from first principles.

Mathematics Coursework (MIT OpenCourseWare)

- **18.01 – Single Variable Calculus** (Completed): Differential and integral calculus with rigorous problem-solving.
- **18.06 – Linear Algebra** (In Progress): Matrix theory, eigendecomposition, orthogonality, and applications to deep learning.

Technical Skills

Languages: Python, C++, SQL
Libraries/Frameworks: NumPy, Pandas, Scikit-Learn, Matplotlib, XGBoost, Transformers, Hugging Face, TensorFlow, PyTorch, OpenCV, DeepFace
Tools & Platforms: Git/GitHub, Docker, REST APIs, Jupyter, VS Code, Firebase, MySQL, CUDA, Obsidian, Zotero, Notion
Typesetting & Writing: LaTeX, Markdown

Certifications

- CS50 AI – Harvard University
- Supervised Machine Learning: Regression and Classification – DeepLearning.AI, Stanford Online
- Advanced Learning Algorithms – DeepLearning.AI, Stanford Online
- Linear Algebra for Machine Learning – DeepLearning.AI
- Calculus for Machine Learning – DeepLearning.AI
- Python Data Analytics – Meta
- Python for Data Science, AI & Development – IBM
- Scientific Computing with Python – FreeCodeCamp.org
- Feature Engineering – Kaggle

Honors & Awards

HackerRank: 5 Stars in Python Programming
HackerRank: 3 Stars in C++ Programming
Kaggle: 17 Badges for contributions and competitions